to a proton, an electron, and an antineutrino—and these nuclei decay at a known rate. By measuring the fraction of carbon-14 that is left in an organism's remains, scientists can determine how long ago the organism died.

In the 1960s physicists developed a theory that described the electromagnetic and weak interactions as aspects of a single *electroweak* interaction. This theory has passed every experimental test to which it has been put. Encouraged by this success, physicists have made similar attempts to describe the strong, electromagnetic, and weak interactions in terms of a single *grand unified theory* (GUT) and have taken steps toward a possible unification of all interactions into a *theory of everything* (TOE). Such theories are still speculative, and there are many unanswered questions in this very active field of current research.

CHAPTER 5 SUMMARY

Using Newton's first law: When an object is in equilibrium in an inertial frame of reference—that is, either at rest or moving with constant velocity—the vector sum of forces acting on it must be zero (Newton's first law). Free-body diagrams are essential in identifying the forces that act on the object being considered.

Newton's third law (action and reaction) is also frequently needed in equilibrium problems. The two forces in an action–reaction pair *never* act on the same object. (See Examples 5.1–5.5.)

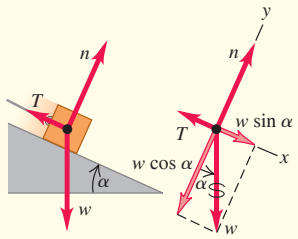
The normal force exerted on an object by a surface is *not* always equal to the object's weight. (See Example 5.4.)

Vector form:

$$\Sigma \vec{F} = \mathbf{0} \qquad (5.1)$$

Component form:

$$\Sigma F_x = 0 \quad \Sigma F_y = 0$$



Using Newton's second law: If the vector sum of forces on an object is *not* zero, the object accelerates. The acceleration is related to the net force by Newton's second law.

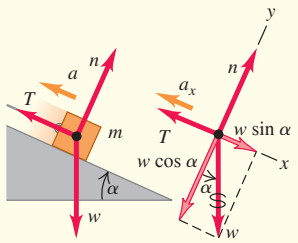
Just as for equilibrium problems, free-body diagrams are essential for solving problems involving Newton's second law, and the normal force exerted on an object is not always equal to its weight. (See Examples 5.6–5.12.)

Vector form:

$$\Sigma \vec{F} = m\vec{a} \qquad (5.2)$$

Component form:

$$\Sigma F_x = ma_x \qquad \Sigma F_y = ma_y$$



Friction and fluid resistance: The contact force between two objects can always be represented in terms of a normal force $\vec{n}$ perpendicular to the surface of contact and a friction force $\vec{f}$ parallel to the surface.

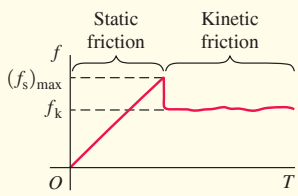
When an object is sliding over the surface, the friction force is called *kinetic* friction. Its magnitude f_k is approximately equal to the normal force magnitude n multiplied by the coefficient of kinetic friction μ_k.

Magnitude of kinetic friction force:

$$f_k = \mu_k n \qquad (5.3)$$

Magnitude of static friction force:

$$f_s \leq (f_s)_{max} = \mu_s n \qquad (5.4)$$



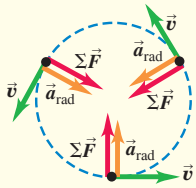
When an object is *not* moving relative to a surface, the friction force is called *static* friction. The *maximum* possible static friction force is approximately equal to the magnitude n of the normal force multiplied by the coefficient of static friction μ_s. The *actual* static friction force may be anything from zero to this maximum value, depending on the situation. Usually μ_s is greater than μ_k for a given pair of surfaces in contact. (See Examples 5.13–5.17.)

Rolling friction is similar to kinetic friction, but the force of fluid resistance depends on the speed of an object through a fluid. (See Example 5.18.)

Forces in circular motion: In uniform circular motion, the acceleration vector is directed toward the center of the circle. The motion is governed by Newton's second law, $\sum\vec{F} = m\vec{a}$. (See Examples 5.19–5.23.)

Acceleration in uniform circular motion:

$$a_{rad} = \frac{v^2}{R} = \frac{4\pi^2 R}{T^2} \qquad (5.13), (5.15)$$



Chapter 5 Media Assets

For assigned homework and other learning materials, go to Mastering Physics.

GUIDED PRACTICE

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review EXAMPLE 5.5 (Section 5.1) before attempting these problems. In all problems, ignore air resistance.

VP5.5.1 In a modified version of the cart and bucket in Fig. 5.5a, the angle of the slope is 36.9° and the bucket weighs 255 N. The cart moves up the incline and the bucket moves downward, both at constant speed. The cable has negligible mass, and there is no friction. (a) What is the weight of the cart? (b) What is the tension in the cable?

VP5.5.2 You increase the angle of the slope in Fig. 5.5a to 25.0° and use a different cart and a different bucket. You observe that the cart and bucket remain at rest when released and that the tension in the cable of negligible mass is 155 N. There is no friction. (a) What is the weight of the cart? (b) What is the *combined* weight of the cart and bucket?

VP5.5.3 You construct a version of the cart and bucket in Fig. 5.5a, but with a slope whose angle can be adjusted. You use a cart of mass 175 kg and a bucket of mass 65.0 kg. The cable has negligible mass, and there is no friction. (a) What must be the angle of the slope so that the cart moves downhill at a constant speed and the bucket moves upward at the same constant speed? (b) With this choice of angle, what will be the tension in the cable?

VP5.5.4 In the situation shown in Fig. 5.5a, let θ be the angle of the slope and suppose there *is* friction between the cart and the track. You find that if the cart and bucket each have the same weight w, they remain at rest when released. In this case, what is the magnitude of the friction force on the cart? Is it less than, greater than, or equal to w?

Be sure to review EXAMPLES 5.13, 5.14, and 5.15 (Section 5.3) before attempting these problems.

VP5.15.1 You pull on a crate using a rope as in Fig. 5.21a, except the rope is at an angle of 20.0° above the horizontal. The weight of the crate is 325 N, and the coefficient of kinetic friction between the crate and the floor is 0.250. (a) What must be the tension in the rope to make the crate move at a constant velocity? (b) What is the normal force that the floor exerts on the crate?

VP5.15.2 You pull on a large box using a rope as in Fig. 5.21a, except the rope is at an angle of 15.0° *below* the horizontal. The weight of the box is 325 N, and the coefficient of kinetic friction between the box and the floor is 0.250. (a) What must be the tension in the rope to make the box move at a constant velocity? (b) What is the normal force that the floor exerts on the box?

VP5.15.3 You are using a lightweight rope to pull a sled along level ground. The sled weighs 475 N, the coefficient of kinetic friction between the sled and the ground is 0.200, the rope is at an angle of 12.0° above the horizontal, and you pull on the rope with a force of 125 N. (a) Find the normal force that the ground exerts on the sled. (b) Find the acceleration of the sled. Is the sled speeding up or slowing down?

VP5.15.4 A large box of mass m sits on a horizontal floor. You attach a lightweight rope to this box, hold the rope at an angle θ above the horizontal, and pull. You find that the minimum tension you can apply to the rope in order to make the box start moving is T_{min}. Find the coefficient of static friction between the floor and the box.

Be sure to review EXAMPLES 5.20, 5.21, and 5.22 (Section 5.4) before attempting these problems.

VP5.22.1 You make a conical pendulum (see Fig. 5.32a) using a string of length 0.800 m and a bob of mass 0.250 kg. When the bob is moving in a circle at a constant speed, the string is at an angle of 20.0° from the vertical. (a) What is the radius of the circle around which the bob moves? (b) How much time does it take the bob to complete one circle? (c) What is the tension in the string?

VP5.22.2 A competition cyclist rides at a constant 12.5 m/s around a curve that is banked at 40.0°. The cyclist and her bicycle have a combined mass of 64.0 kg. (a) What must be the radius of her turn if there is to be no friction force pushing her either up or down the banked curve? (b) What is the magnitude of her acceleration? (c) What is the magnitude of the normal force that the surface of the banked curve exerts on the bicycle?

VP5.22.3 An aerobatic airplane flying at a constant 80.0 m/s makes a horizontal turn of radius 175 m. The pilot has mass 80.0 kg. (a) What is the bank angle of the airplane? (b) What is the pilot's apparent weight during the turn? How many times greater than his actual weight is this?

VP5.22.4 A sports car moves around a banked curve at just the right constant speed v so that no friction is needed to make the turn. During the turn, the driver (mass m) feels as though she weighs x times her actual weight. (a) Find the magnitude of the *net* force on the driver during the turn in terms of m, g, and x. (b) Find the radius of the turn in terms of v, g, and x.

BRIDGING PROBLEM In a Rotating Cone

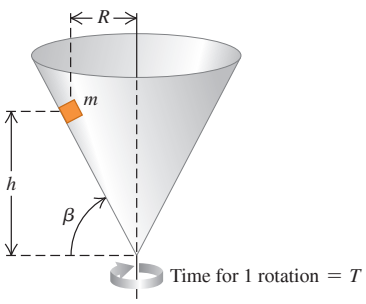
A small block with mass m is placed inside an inverted cone that is rotating about a vertical axis such that the time for one revolution of the cone is T (**Fig. 5.39**). The walls of the cone make an angle β with the horizontal. The coefficient of static friction between the block and the cone is μ_s. If the block is to remain at a constant height h above the apex of the cone, what are (a) the maximum value of T and (b) the minimum value of T? (That is, find expressions for T_{max} and T_{min} in terms of β and h.)

SOLUTION GUIDE

IDENTIFY and SET UP

1. Although we want the block not to slide up or down on the inside of the cone, this is *not* an equilibrium problem. The block rotates with the cone and is in uniform circular motion, so it has an acceleration directed toward the center of its circular path.
2. Identify the forces on the block. What is the direction of the friction force when the cone is rotating as slowly as possible, so T has its maximum value T_{max}? What is the direction of the friction force when the cone is rotating as rapidly as possible, so T has its minimum value T_{min}? In these situations does the static friction force have its *maximum* magnitude? Why or why not?
3. Draw a free-body diagram for the block when the cone is rotating with $T = T_{max}$ and a free-body diagram when the cone is rotating with $T = T_{min}$. Choose coordinate axes, and remember that it's usually easiest to choose one of the axes to be in the direction of the acceleration.
4. What is the radius of the circular path that the block follows? Express this in terms of β and h.
5. List the unknown quantities, and decide which of these are the target variables.

Figure 5.39 A block inside a spinning cone.



EXECUTE

6. Write Newton's second law in component form for the case in which the cone is rotating with $T = T_{max}$. Write the acceleration in terms of T_{max}, β, and h, and write the static friction force in terms of the normal force n.
7. Solve these equations for the target variable T_{max}.
8. Repeat steps 6 and 7 for the case in which the cone is rotating with $T = T_{min}$, and solve for the target variable T_{min}.

EVALUATE

9. You'll end up with some fairly complicated expressions for T_{max} and T_{min}, so check them over carefully. Do they have the correct units? Is the minimum time T_{min} less than the maximum time T_{max}, as it must be?
10. What do your expressions for T_{max} and T_{min} become if $\mu_s = 0$? Check your results by comparing them with Example 5.22 in Section 5.4.

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

[Always assume that pulleys are frictionless and massless and that strings and cords are massless, unless otherwise noted.]

DISCUSSION QUESTIONS

Q5.1 A man sits in a seat that is hanging from a rope. The rope passes over a pulley suspended from the ceiling, and the man holds the other end of the rope in his hands. What is the tension in the rope, and what force does the seat exert on him? Draw a free-body force diagram for the man.

Q5.2 "In general, the normal force is not equal to the weight." Give an example in which these two forces are equal in magnitude, and at least two examples in which they are not.

Q5.3 A clothesline hangs between two poles. No matter how tightly the line is stretched, it sags a little at the center. Explain why.

Q5.4 You drive a car up a steep hill at constant speed. Discuss all of the forces that act on the car. What pushes it up the hill?

Q5.5 For medical reasons, astronauts in outer space must determine their body mass at regular intervals. Devise a scheme for measuring body mass in an apparently weightless environment.

Q5.6 To push a box up a ramp, which requires less force: pushing horizontally or pushing parallel to the ramp? Why?

Q5.7 A woman in a lift lets go of her briefcase, but it does not fall to the floor. How is the lift moving?

Q5.8 A block rests on an inclined plane with enough friction to prevent it from sliding down. To start the block moving, is it easier to push it up the plane or down the plane? Why?

Q5.9 A crate slides up an inclined ramp and then slides down the ramp after momentarily stopping near the top. There is kinetic friction between the surface of the ramp and the crate. Which is greater? (i) The crate's acceleration going up the ramp; (ii) the crate's acceleration going down the ramp; (iii) both are the same. Explain.

Q5.10 A crate of books rests on a level floor. To move it along the floor at a constant velocity, why do you exert less force if you pull it at an angle θ above the horizontal than if you push it at the same angle below the horizontal?

Q5.11 In a world without friction, which of the following activities could you do (or not do)? Explain your reasoning. (a) Drive around an unbanked highway curve; (b) jump into the air; (c) start walking on a horizontal footpath; (d) climb a vertical ladder; (e) change lanes while you drive.

Q5.12 When you stand with bare feet in a wet bathtub, the grip feels fairly secure, and yet a catastrophic slip is quite possible. Explain this in terms of the two coefficients of friction.

Q5.13 You are pushing a large crate from the back of a freight lift to the front as the lift is moving to the next floor. In which situation is the force you must apply to move the crate the least, and in which is it the greatest: when the lift is accelerating upward, when it is accelerating downward, or when it is traveling at constant speed? Explain.

Q5.14 It is often said that "friction always opposes motion." Give at least one example in which (a) static friction *causes* motion, and (b) kinetic friction *causes* motion.

Q5.15 If there is a net force on a particle in uniform circular motion, why doesn't the particle's speed change?

Q5.16 A curve in a road has a bank angle calculated and posted for 80 km/h. However, the road is covered with ice, so you cautiously plan to drive slower than this limit. What might happen to your car? Why?

Q5.17 You swing a ball on the end of a lightweight string in a horizontal circle at constant speed. Can the string ever be truly horizontal? If not, would it slope above the horizontal or below the horizontal? Why?

Q5.18 The centrifugal force is not included in the free-body diagrams of Figs. 5.34b and 5.35. Explain why not.

Q5.19 A professor swings a rubber stopper in a horizontal circle on the end of a string in front of his class. He tells Caroline, in the front row, that he is going to let the string go when the stopper is directly in front of her face. Should Caroline worry?

Q5.20 To keep the forces on the riders within allowable limits, many loop-the-loop roller coaster rides are designed so that the loop is not a perfect circle but instead has a larger radius of curvature at the bottom than at the top. Explain.

Q5.21 A tennis ball drops from rest at the top of a tall glass cylinder—first with the air pumped out of the cylinder so that there is no air resistance, and again after the air has been readmitted to the cylinder. You examine multiflash photographs of the two drops. Can you tell which photo belongs to which drop? If so, how?

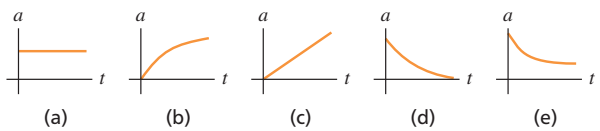
Q5.22 You throw a tennis ball straight upward with speed v_0. When the ball returns to the point from where you threw it, how does its speed compare to v_0 (a) in the absence of air resistance and (b) in the presence of air resistance? Explain.

Q5.23 You throw a tennis ball straight upward. If you do *not* ignore air resistance, how does the time required for the ball to reach its maximum height compare to the time required for it to fall from its maximum height back down to the height from which you threw it? Explain.

Q5.24 You have two identical tennis balls and fill one with water. You release both balls simultaneously from the top of a tall building. If air resistance is negligible, which ball will strike the ground first? Explain. What if air resistance is *not* negligible?

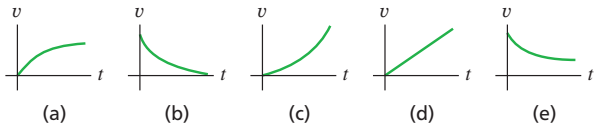
Q5.25 A ball is dropped from rest and feels air resistance as it falls. Which of the graphs in **Fig. Q5.25** best represents its acceleration as a function of time?

Figure **Q5.25**



Q5.26 A ball is dropped from rest and feels air resistance as it falls. Which of the graphs in **Fig. Q5.26** best represents its vertical velocity component as a function of time, if the $+y$-direction is taken to be downward?

Figure **Q5.26**



Q5.27 When a batted baseball moves with air drag, when does the ball travel a greater horizontal distance? (i) While climbing to its maximum height; (ii) while descending from its maximum height back to the ground; (iii) the same for both. Explain in terms of the forces acting on the ball.

Q5.28 "A ball is thrown from the edge of a high cliff. Regardless of the angle at which it is thrown, due to air resistance, the ball will eventually end up moving vertically downward." Justify this statement.

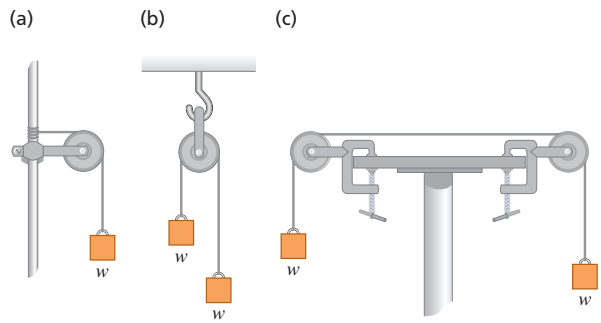
EXERCISES

Section 5.1 Using Newton's First Law: Particles in Equilibrium

5.1 • Two 23.0-N weights are suspended at opposite ends of a rope that passes over a light, frictionless pulley. The pulley is attached to a chain from the ceiling. (a) What is the tension in the rope? (b) What is the tension in the chain?

5.2 • In **Fig. E5.2** each of the suspended blocks has weight w. The pulleys are frictionless, and the ropes have negligible weight. In each case, draw a free-body diagram and calculate the tension T in the rope in terms of w.

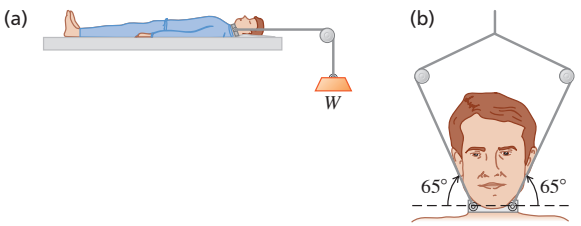
Figure **E5.2**



5.3 • A 75.0 kg wrecking ball hangs from a uniform, heavy-duty chain of mass 26.0 kg. (a) Find the maximum and minimum tensions in the chain. (b) What is the tension at a point three-fourths of the way up from the bottom of the chain?

5.4 •• BIO **Injuries to the Spinal Column.** In the treatment of spine injuries, it is often necessary to provide tension along the spinal column to stretch the backbone. One device for doing this is the Stryker frame (**Fig. E5.4a**). A weight W is attached to the patient (sometimes around a neck collar, Fig. E5.4b), and friction between the person's body and the bed prevents sliding. (a) If the coefficient of static friction between a 80.5-kg patient's body and the bed is 0.745, what is the maximum traction force along the spinal column that W can provide without causing the patient to slide? (b) Under the conditions of maximum traction, what is the tension in each cable attached to the neck collar?

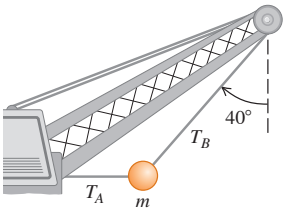
Figure E5.4



5.5 •• A picture frame hung against a wall is suspended by two wires attached to its upper corners. If the two wires make the same angle with the vertical, what must this angle be if the tension in each wire is equal to 0.75 of the weight of the frame? (Ignore any friction between the wall and the picture frame.)

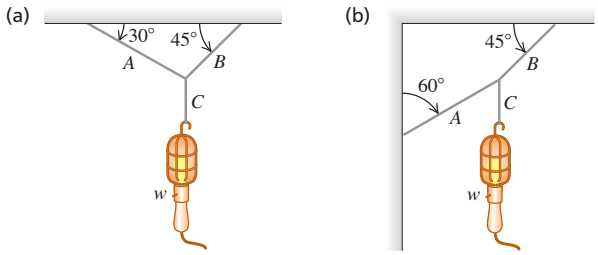
5.6 •• A large wrecking ball is held in place by two light steel cables (**Fig. E5.6**). If the mass m of the wrecking ball is 3620 kg, what are (a) the tension T_B in the cable that makes an angle of 40° with the vertical and (b) the tension T_A in the horizontal cable?

Figure E5.6



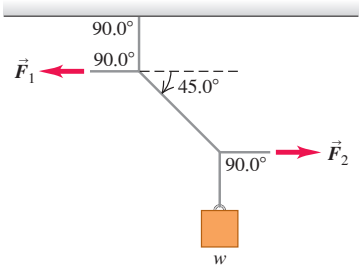
5.7 •• Find the tension in each cord in **Fig. E5.7** if the weight of the suspended object is w.

Figure E5.7



5.9 •• In **Fig. E5.8** the weight w is 60.0 N. (a) What is the tension in the diagonal string? (b) Find the magnitudes of the horizontalforces $\vec{F}_1$ and $\vec{F}_2$ that must be applied to hold the system in the position shown.

Figure E5.8



5.9 •• A man pushes on a piano with mass 180 kg; it slides at constant velocity down a ramp that is inclined at 18.4° above the horizontal floor. Neglect any friction acting on the piano. Calculate the magnitude of the force applied by the man if he pushes (a) parallel to the incline and (b) parallel to the floor.

Section 5.2 Using Newton's Second Law: Dynamics of Particles

5.10 •• **Apparent Weight.** A 505 N physics student stands on a bathroom scale in a lift that is supported by a cable. The combined mass of student plus lift is 863 kg. As the lift starts moving, the scale reads 477 N. (a) Find the acceleration of the lift (magnitude and direction). (b) What is the acceleration if the scale reads 688 N? (c) If the scale reads zero, should the student worry? Explain. (d) What is the tension in the cable in parts (a) and (c)?

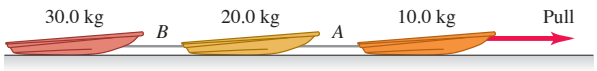
5.11 •• BIO **Stay Awake!** An astronaut is inside a 2.25×10^6 kg rocket that is blasting off vertically from the launch pad. You want this rocket to reach the speed of sound (331 m/s) as quickly as possible, but astronauts are in danger of blacking out at an acceleration greater than $4g$. (a) What is the maximum initial thrust this rocket's engines can have but just barely avoid blackout? Start with a free-body diagram of the rocket. (b) What force, in terms of the astronaut's weight w, does the rocket exert on her? Start with a free-body diagram of the astronaut. (c) What is the shortest time it can take the rocket to reach the speed of sound?

5.12 •• A rocket of initial mass 125 kg (including all the contents) has an engine that produces a constant vertical force (the *thrust*) of 1720 N. Inside this rocket, a 15.5 N electric power supply rests on the floor. (a) Find the initial acceleration of the rocket. (b) When the rocket initially accelerates, how hard does the floor push on the power supply? (*Hint:* Start with a free-body diagram for the power supply.)

5.13 •• CP ***Genesis* Crash.** On September 8, 2004, the *Genesis* spacecraft crashed in the Utah desert because its parachute did not open. The 210 kg capsule hit the ground at 311 km/h and penetrated the soil to a depth of 81.0 cm. (a) What was its acceleration (in m/s^2 and in g's), assumed to be constant, during the crash? (b) What force did the ground exert on the capsule during the crash? Express the force in newtons and as a multiple of the capsule's weight. (c) How long did this force last?

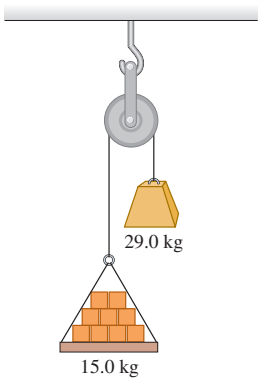
5.14 • Three sleds are being pulled horizontally on frictionless horizontal ice using horizontal ropes (**Fig. E5.14**). The pull is of magnitude 132 N. Find (a) the acceleration of the system and (b) the tension in ropes A and B.

Figure E5.14



5.15 •• **Atwood's Machine.** A 15.0 kg load of bricks hangs from one end of a rope that passes over a small, frictionless pulley. A 29.0 kg counterweight is suspended from the other end of the rope (**Fig. E5.15**). The system is released from rest. (a) Draw two free-body diagrams, one for the load of bricks and one for the counterweight. (b) What is the magnitude of the upward acceleration of the load of bricks? (c) What is the tension in the rope while the load is moving? How does the tension compare to the weight of the load of bricks? To the weight of the counterweight?

Figure E5.15



5.16 •• **CP** An 8.00 kg block of ice, released from rest at the top of a 1.31-m-long frictionless ramp, slides downhill, reaching a speed of 2.59 m/s at the bottom. (a) What is the angle between the ramp and the horizontal? (b) What would be the speed of the ice at the bottom if the motion were opposed by a constant friction force of 10.0 N parallel to the surface of the ramp?

5.17 •• A light rope is attached to a block with mass 4.70 kg that rests on a frictionless, horizontal surface. The horizontal rope passes over a frictionless, massless pulley, and a block with mass m is suspended from the other end. When the blocks are released, the tension in the rope is 13.6 N. (a) Draw two free-body diagrams: one for each block. (b) What is the acceleration of either block? (c) Find m. (d) How does the tension compare to the weight of the hanging block?

5.18 •• **CP** **Runway Design.** A transport plane takes off from a level landing field with two gliders in tow, one behind the other. The mass of each glider is 700 kg, and the total resistance (air drag plus friction with the runway) on each may be assumed constant and equal to 2700 N. The tension in the towrope between the transport plane and the first glider is not to exceed 12,000 N. (a) If a speed of 40 m/s is required for takeoff, what minimum length of runway is needed? (b) What is the tension in the towrope between the two gliders while they are accelerating for the takeoff?

5.19 •• **CP** **BIO** **Force During a Jump.** When jumping straight up from a crouched position, an average person can reach a maximum height of about 60 cm. During the jump, the person's body from the knees up typically rises a distance of around 50 cm. To keep the calculations simple and yet get a reasonable result, assume that the *entire body* rises this much during the jump. (a) With what initial speed does the person leave the ground to reach a height of 60 cm? (b) Draw a free-body diagram of the person during the jump. (c) In terms of this jumper's weight w, what force does the ground exert on him or her during the jump?

5.20 **CP** **CALC** A 2080 kg test rocket is launched vertically from the launch pad. Its fuel (of negligible mass) provides a thrust force such that its vertical velocity as a function of time is given by $v(t) = At + Bt^2$, where A and B are constants and time is measured from the instant the fuel is ignited. The rocket has an upward acceleration of 1.80 m/s^2 at the instant of ignition and, 1.10 s later, an upward velocity of 1.80 m/s. (a) Determine A and B, including their SI units. (b) At 4.00 s after fuel ignition, what is the acceleration of the rocket, and (c) what thrust force does the burning fuel exert on it, assuming no air resistance? Express the thrust in newtons and as a multiple of the rocket's weight. (d) What was the initial thrust due to the fuel?

5.21 •• **CP** **CALC** A 2.10 kg box is moving to the right with speed 8.50 m/s on a horizontal, frictionless surface. At $t = 0$ a horizontal force is applied to the box. The force is directed to the left and has magnitude $F(t) = (6.00\ \text{N/s}^2)t^2$. (a) What distance does the box move from its position at $t = 0$ before its speed is reduced to zero? (b) If the force continues to be applied, what is the speed of the box at $t = 3.00$ s?

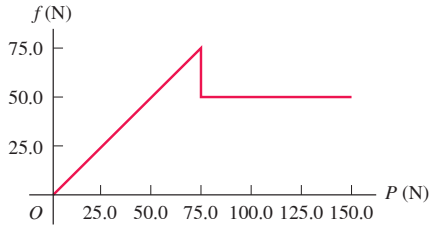
5.22 •• **CP** **CALC** A 4.50 kg crate is suspended from the end of a short vertical rope of negligible mass. An upward force $F(t)$ is applied to the end of the rope, and the height of the crate above its initial position is given by $y(t) = (2.80\ \text{m/s})t + (0.610\ \text{m/s}^3)t^3$. What is the magnitude of F when $t = 4.30$ s?

SECTION 5.3 Friction Forces

5.23 • **BIO** **The Trendelenburg Position.** After emergencies with major blood loss, a patient is placed in the Trendelenburg position, in which the foot of the bed is raised to get maximum blood flow to the brain. If the coefficient of static friction between a typical patient and the bedsheets is 1.20, what is the maximum angle at which the bed can be tilted with respect to the floor before the patient begins to slide?

5.24 • In a laboratory experiment on friction, a 130 N block resting on a rough horizontal table is pulled by a horizontal wire. The pull gradually increases until the block begins to move and continues to increase thereafter. **Figure E5.24** shows a graph of the friction force on this block as a function of the pull. (a) Identify the regions of the graph where static friction and kinetic friction occur. (b) Find the coefficients of static friction and kinetic friction between the block and the table. (c) Why does the graph slant upward at first but then level out? (d) What would the graph look like if a 130 N brick were placed on the block, and what would the coefficients of friction be?

Figure **E5.24**



5.25 •• **CP** A stockroom worker pushes a box with mass 16.8 kg on a horizontal surface with a constant speed of 3.50 m/s. The coefficient of kinetic friction between the box and the surface is 0.20. (a) What horizontal force must the worker apply to maintain the motion? (b) If the force calculated in part (a) is removed, how far does the box slide before coming to rest?

5.26 •• A box of bananas weighing 40.0 N rests on a horizontal surface. The coefficient of static friction between the box and the surface is 0.40, and the coefficient of kinetic friction is 0.20. (a) If no horizontal force is applied to the box and the box is at rest, how large is the friction force exerted on it? (b) What is the magnitude of the friction force if a monkey applies a horizontal force of 6.0 N to the box and the box is initially at rest? (c) What minimum horizontal force must the monkey apply to start the box in motion? (d) What minimum horizontal force must the monkey apply to keep the box moving at constant velocity once it has been started? (e) If the monkey applies a horizontal force of 18.0 N, what is the magnitude of the friction force and what is the box's acceleration?

5.27 •• A 38.4 kg crate of tools rests on a horizontal floor. You exert a gradually increasing horizontal push on it, and the crate just begins to move when your force exceeds 320 N. Then you must reduce your push to 221 N to keep it moving at a steady 24.8 cm/s. (a) What are the coefficients of static and kinetic friction between the crate and the floor? (b) What push must you exert to give it an acceleration of 1.41 m/s^2? (c) Suppose you were performing the same experiment on the moon, where the acceleration due to gravity is 1.62 m/s^2. (i) What magnitude push would cause it to move? (ii) What would its acceleration be if you maintained the push in part (b)?

5.28 • Consider the heaviest box that you can push at constant speed across a level floor, where the coefficient of kinetic friction is 0.50, and estimate the maximum horizontal force that you can apply to the box. A box sits on a ramp that is inclined at an angle of 60° above the horizontal. The coefficient of kinetic friction between the box and the ramp is 0.50. If you apply the same magnitude force, now parallel to the ramp, that you applied to the box on the floor, what is the heaviest box that you can push up the ramp at constant speed? (In both cases assume you can give enough extra push to get the box started moving.)

5.29 • **CP** Estimate the height of a typical playground slide and the angle its surface makes with the horizontal. (a) Some children like to slide down while sitting on a sheet of wax paper. This makes the friction force exerted by the slide very small. If a child starts from rest and we take the friction force to be zero, what is the speed of the child when he reaches the bottom of the slide? (b) If the child doesn't use the wax paper, his speed at the bottom is half the value calculated in part (a). What is the coefficient of kinetic friction between the child and the slide when wax paper isn't used? (c) A child wearing a different sort of clothing than the first child climbs the ladder to the top of the slide, sits on the slide, lets go of the handrail, and remains at rest. What is the minimum possible value for the coefficient of static friction between this child and the surface of the slide?

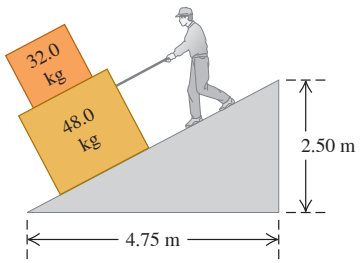
5.30 •• Some sliding rocks approach the base of a hill with a speed of 13 m/s. The hill rises at 42° above the horizontal and has coefficients of kinetic friction and static friction of 0.43 and 0.61, respectively, with these rocks. (a) Find the acceleration of the rocks as they slide up the hill. (b) Once a rock reaches its highest point, will it stay there or slide down the hill? If it stays, show why. If it slides, find its acceleration on the way down.

5.31 •• A box with mass 10.0 kg moves on a ramp that is inclined at an angle of 55.0° above the horizontal. The coefficient of kinetic friction between the box and the ramp surface is $\mu_k = 0.300$. Calculate the magnitude of the acceleration of the box if you push on the box with a constant force $F = 120.0$ N that is parallel to the ramp surface and (a) directed down the ramp, moving the box down the ramp; (b) directed up the ramp, moving the box up the ramp.

5.32 •• A pickup truck is carrying a toolbox, but the rear gate of the truck is missing. The toolbox will slide out if it is set moving. The coefficients of kinetic friction and static friction between the box and the level bed of the truck are 0.355 and 0.650, respectively. Starting from rest, what is the shortest time this truck could accelerate uniformly to 30.0 m/s without causing the box to slide? Draw a free-body diagram of the toolbox.

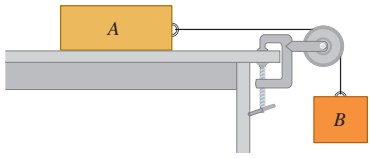
5.33 •• You are lowering two boxes, one on top of the other, down a ramp by pulling on a rope parallel to the surface of the ramp (**Fig. E5.33**). Both boxes move together at a constant speed of 15.0 cm/s. The coefficient of kinetic friction between the ramp and the lower box is 0.444, and the coefficient of static friction between the two boxes is 0.800. (a) What force do you need to exert to accomplish this? (b) What are the magnitude and direction of the friction force on the upper box?

Figure **E5.33**



5.34 •• Consider the system shown in **Fig. E5.34.** Block A weighs 45.0 N, and block B weighs 25.0 N. Once block B is set into downward motion, it descends at a constant speed. (a) Calculate the coefficient of kinetic friction between block A and the tabletop. (b) A cat, also of weight 45.0 N, falls asleep on top of block A. If block B is now set into downward motion, what is its acceleration (magnitude and direction)?

Figure **E5.34**

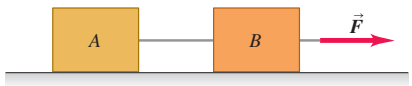


5.35 •• **CP** **Stopping Distance.** (a) If the coefficient of kinetic friction between tires and dry pavement is 0.80, what is the shortest distance in which you can stop a car by locking the brakes when the car is traveling at 28.7 m/s? (b) On wet pavement the coefficient of kinetic friction may be only 0.25. How fast should you drive on wet pavement to be able to stop in the same distance as in part (a)? (*Note:* Locking the brakes is *not* the safest way to stop.)

5.36 •• **CP** A 25.4 kg box of textbooks rests on a loading ramp that makes an angle α with the horizontal. The coefficient of kinetic friction is 0.25, and the coefficient of static friction is 0.37. (a) As α is increased, find the minimum angle at which the box starts to slip. (b) At this angle, find the acceleration once the box has begun to move. (c) At this angle, how fast will the box be moving after it has slid 5.3 m along the loading ramp?

5.37 • Two crates connected by a rope lie on a horizontal surface (**Fig. E5.37**). Crate A has mass m_A, and crate B has mass m_B. The coefficient of kinetic friction between each crate and the surface is μ_k. The crates are pulled to the right at constant velocity by a horizontal force $\vec{F}$. Draw one or more free-body diagrams to calculate the following in terms of m_A, m_B, and μ_k: (a) the magnitude of $\vec{F}$ and (b) the tension in the rope connecting the blocks.

Figure **E5.37**



5.38 •• A box with mass m is dragged across a level floor with coefficient of kinetic friction μ_k by a rope that is pulled upward at an angle θ above the horizontal with a force of magnitude F. (a) In terms of m, μ_k, θ, and g, obtain an expression for the magnitude of the force required to move the box with constant speed. (b) Knowing that you are studying physics, a CPR instructor asks you how much force it would take to slide a 90 kg patient across a floor at constant speed by pulling on him at an angle of 25° above the horizontal. By dragging weights wrapped in an old pair of pants down the hall with a spring balance, you find that $\mu_k = 0.35$. Use the result of part (a) to answer the instructor's question.

5.39 •• **CP** As shown in Fig. E5.34, block A (mass 2.19 kg) rests on a tabletop. It is connected by a horizontal cord passing over a light, frictionless pulley to a hanging block B (mass 2.27 kg). The coefficient of kinetic friction between block A and the tabletop is 0.331. The blocks are released then from rest. Draw one or more free-body diagrams to find (a) the speed of each block after they move 3.30 cm and (b) the tension in the cord.

5.40 •• You throw a cricket ball straight upward. The drag force is proportional to v^2. In terms of g, what is the y-component of the ball's acceleration when the ball's speed is half its terminal speed and (a) it is moving up? (b) It is moving back down?

5.41 •• A large crate with mass m rests on a horizontal floor. The coefficients of friction between the crate and the floor are μ_s and μ_k. A woman pushes downward with a force $\vec{F}$ on the crate at an angle θ below the horizontal. (a) What magnitude of force $\vec{F}$ is required to keep the crate moving at constant velocity? (b) If μ_s is greater than some critical value, the woman cannot start the crate moving no matter how hard she pushes. Calculate this critical value of μ_s.

Section 5.4 Dynamics of Circular Motion

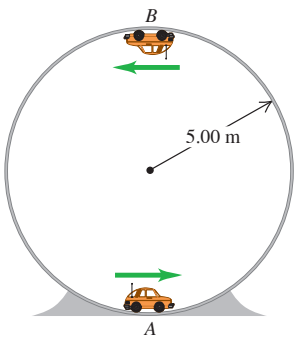
5.42 • You are sitting on the edge of a horizontal disk (for example, a playground merry-go-round) that has radius 3.00 m and is rotating at a constant rate about a vertical axis. (a) If the coefficient of static friction between you and the surface of the disk is 0.400, what is the minimum time for one revolution of the disk if you are not to slide off? (b) Your friend's weight is half yours. If the coefficient of static friction for him is the same as for you, what is the minimum time for one revolution if he is not to slide off?

5.43 • A stone with mass 0.80 kg is attached to one end of a string 0.90 m long. The string will break if its tension exceeds 60.0 N. The stone is whirled in a horizontal circle on a frictionless tabletop; the other end of the string remains fixed. (a) Draw a free-body diagram of the stone. (b) Find the maximum speed the stone can attain without the string breaking.

5.44 • BIO **Force on a Skater's Wrist.** A 52 kg ice skater spins about a vertical axis through her body with her arms horizontally outstretched; she makes 2.0 turns each second. The distance from one hand to the other is 1.50 m. Biometric measurements indicate that each hand typically makes up about 1.25% of body weight. (a) Draw a free-body diagram of one of the skater's hands. (b) What horizontal force must her wrist exert on her hand? (c) Express the force in part (b) as a multiple of the weight of her hand.

5.45 •• A small remote-controlled car with mass 1.60 kg moves at a constant speed of $v = 12.0$ m/s in a track formed by a vertical circle inside a hollow metal cylinder that has a radius of 5.00 m (**Fig. E5.45**). What is the magnitude of the normal force exerted on the car by the walls of the cylinder at (a) point A (bottom of the track) and (b) point B (top of the track)?

Figure E5.45



5.46 •• A small car with mass 0.800 kg travels at constant speed on the inside of a track that is a vertical circle with radius 5.00 m (Fig. E5.45). If the normal force exerted by the track on the car when it is at the top of the track (point B) is 6.00 N, what is the normal force on the car when it is at the bottom of the track (point A)?

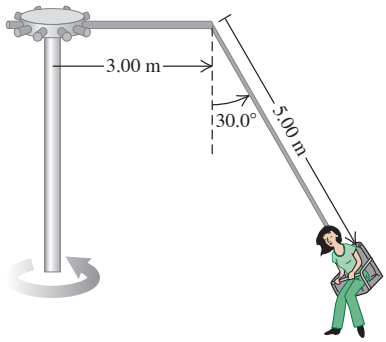
5.47 • A small model car with mass m travels at constant speed on the inside of a track that is a vertical circle with radius 5.00 m (Fig. E5.45). If the normal force exerted by the track on the car when it is at the bottom of the track (point A) is equal to $2.50mg$, how much time does it take the car to complete one revolution around the track?

5.48 • A flat (unbanked) curve on a highway has a radius of 230.0 m. A car rounds the curve at a speed of 28.0 m/s. (a) What is the minimum coefficient of static friction that will prevent sliding? (b) Suppose that the highway is icy and the coefficient of static friction between the tires and pavement is only one-third of what you found in part (a). What should be the maximum speed of the car so that it can round the curve safely?

5.49 •• A 1125 kg car and a 2250 kg pickup truck approach a curve on a highway that has a radius of 225 m. (a) At what angle should the highway engineer bank this curve so that vehicles traveling at 105 km/h can safely round it regardless of the condition of their tires? Should the heavy truck go slower than the lighter car? (b) As the car and truck round the curve at 105 km/h, find the normal force on each one due to the highway surface.

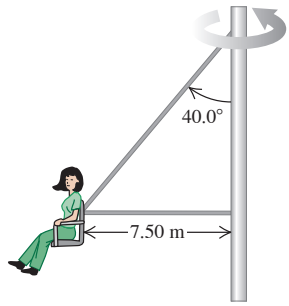
5.50 •• The "Giant Swing" at a county fair consists of a vertical central shaft with a number of horizontal arms attached at its upper end. Each arm supports a seat suspended from a cable 5.00 m long, and the upper end of the cable is fastened to the arm at a point 3.00 m from the central shaft (**Fig. E5.50**). (a) Find the time of one revolution of the swing if the cable supporting a seat makes an angle of 30.0° with the vertical. (b) Does the angle depend on the weight of the passenger for a given rate of revolution?

Figure E5.50



5.51 •• In another version of the "Giant Swing" (see Exercise 5.50), the seat is connected to two cables, one of which is horizontal (**Fig. E5.51**). The seat swings in a horizontal circle at a rate of 28.0 rpm (rev/min). If the seat weighs 255 N and an 825 N person is sitting in it, find the tension in each cable.

Figure E5.51



5.52 • A steel ball with mass m is suspended from the ceiling at the bottom end of a light, 15.0-m-long rope. The ball swings back and forth like a pendulum. When the ball is at its lowest point and the rope is vertical, the tension in the rope is three times the weight of the ball, so $T = 3mg$. (a) What is the speed of the ball as it swings through this point? (b) What is the speed of the ball if $T = mg$ at this point, where the rope is vertical?

5.53 •• **Rotating Space Stations.** One problem for humans living in outer space is that they are apparently weightless. One way around this problem is to design a space station that spins about its center at a constant rate. This creates "artificial gravity" at the outside rim of the station. (a) If the diameter of the space station is 890 m, how many revolutions per minute are needed for the "artificial gravity" acceleration to be 9.80 m/s^2? (b) If the space station is a waiting area for travelers going to Mars, it might be desirable to simulate the acceleration due to gravity on the Martian surface (3.70 m/s^2). How many revolutions per minute are needed in this case?

5.54 • The Cosmo Clock 21 Ferris wheel in Yokohama, Japan, has a diameter of 100 m. Its name comes from its 60 arms, each of which can function as a second hand (so that it makes one revolution every 60.0 s). (a) Find the speed of the passengers when the Ferris wheel is rotating at this rate. (b) A passenger weighs 902 N at the weight-guessing booth on the ground. What is his apparent weight at the highest and at the lowest point on the Ferris wheel? (c) What would be the time for one revolution if the passenger's apparent weight at the highest point were zero? (d) What then would be the passenger's apparent weight at the lowest point?

5.55 •• A small rock with mass m is attached to a light string of length L and whirled in a vertical circle of radius R. (a) What is the minimum speed v at the rock's highest point for which it stays in a circular path? (b) If the speed at the rock's lowest point in its circular path is twice the value found in part (a), what is the tension in the string when the rock is at this point?

5.56 •• A 50.0 kg stunt pilot who has been diving her airplane vertically pulls out of the dive by changing her course to a circle in a vertical plane. (a) If the plane's speed at the lowest point of the circle is 95.0 m/s, what is the minimum radius of the circle so that the acceleration at this point will not exceed 4.00g? (b) What is the apparent weight of the pilot at the lowest point of the pullout?

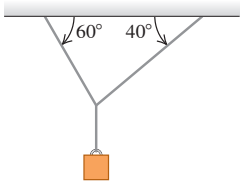
5.57 • **Stay Dry!** You tie a cord to a pail of water and swing the pail in a vertical circle of radius 0.600 m. What minimum speed must you give the pail at the highest point of the circle to avoid spilling water?

5.58 •• A bowling ball weighing 71.2 N is attached to the ceiling by a 3.80 m rope. The ball is pulled to one side and released; it then swings back and forth as a pendulum. As the rope swings through the vertical, the speed of the bowling ball is 4.20 m/s. At this instant, what are (a) the acceleration of the bowling ball, in magnitude and direction, and (b) the tension in the rope?

PROBLEMS

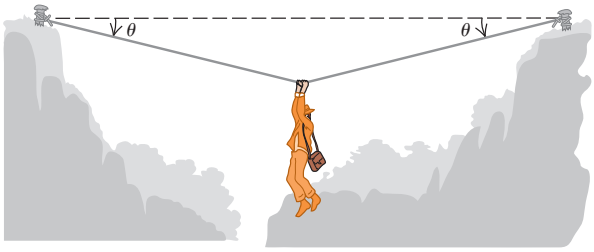
5.59 ••• Two ropes are connected to a steel cable that supports a hanging weight (**Fig. P5.59**). (a) Draw a free-body diagram showing all of the forces acting at the knot that connects the two ropes to the steel cable. Based on your diagram, which of the two ropes will have the greater tension? (b) If the maximum tension either rope can sustain without breaking is 5000 N, determine the maximum value of the hanging weight that these ropes can safely support. Ignore the weight of the ropes and of the steel cable.

Figure P5.59



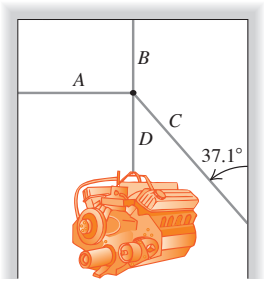
5.60 •• An adventurous archaeologist crosses between two rock cliffs by slowly going hand over hand along a rope stretched between the cliffs. He stops to rest at the middle of the rope (**Fig. P5.60**). The rope will break if the tension in it exceeds 2.10×10^4 N, and our hero's mass is 91.2 kg. (a) If the angle θ is 10.0°, what is the tension in the rope? (b) What is the smallest value θ can have if the rope is not to break?

Figure P5.60



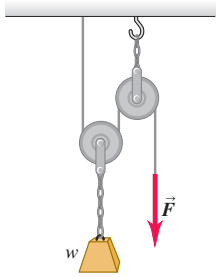
5.61 •• In a repair shop a truck engine that has mass 409 kg is held in place by four light cables (**Fig. P5.61**). Cable A is horizontal, cables B and D are vertical, and cable C makes an angle of 37.1° with a vertical wall. If the tension in cable A is 722 N, what are the tensions in cables B and C?

Figure P5.61



5.62 •• In **Fig. P5.62** a worker lifts a weight w by pulling down on a rope with a force $\vec{F}$. The upper pulley is attached to the ceiling by a chain, and the lower pulley is attached to the weight by another chain. Draw one or more free-body diagrams to find the tension in each chain and the magnitude of $\vec{F}$, in terms of w, if the weight is lifted at constant speed. Assume that the rope, pulleys, and chains have negligible weights.

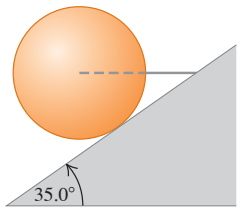
Figure P5.62



5.63 •• **CP** A small block sits at one end of a flat board that is 3.00 m long. The coefficients of friction between the block and the board are $\mu_s = 0.600$ and $\mu_k = 0.400$. The end of the board where the block sits is slowly raised until the angle the board makes with the horizontal is α_0, and then the block starts to slide down the board. If the angle is kept equal to α_0 as the block slides, what is the speed of the block when it reaches the bottom of the board?

5.64 ••• A horizontal wire holds a solid uniform ball of mass m in place on a tilted ramp that rises 35.0° above the horizontal. The surface of this ramp is perfectly smooth, and the wire is directed away from the center of the ball (**Fig. P5.64**). (a) Draw a free-body diagram of the ball. (b) How hard does the surface of the ramp push on the ball? (c) What is the tension in the wire?

Figure P5.64



5.65 •• **CP** A box of mass 12.0 kg sits at rest on a horizontal surface. The coefficient of kinetic friction between the surface and the box is 0.300. The box is initially at rest, and then a constant force of magnitude F and direction 37.0° below the horizontal is applied to the box; the box slides along the surface. (a) What is F if the box has a speed of 6.00 m/s after traveling a distance of 8.00 m? (b) What is F if the surface is frictionless and all the other quantities are the same? (c) What is F if all the quantities are the same as in part (a) but the force applied to the box is horizontal?

5.66 •• **CP** A box is sliding with a constant speed of 4.00 m/s in the $+x$-direction on a horizontal, frictionless surface. At $x = 0$ the box encounters a rough patch of the surface, and then the surface becomes even rougher. Between $x = 0$ and $x = 2.00$ m, the coefficient of kinetic friction between the box and the surface is 0.200; between $x = 2.00$ m and $x = 4.00$ m, it is 0.400. (a) What is the x-coordinate of the point where the box comes to rest? (b) How much time does it take the box to come to rest after it first encounters the rough patch at $x = 0$?

5.67 •• **CP** **BIO** **Forces During Chin-ups.** When you do a chin-up, you raise your chin just over a bar (the chinning bar), supporting yourself with only your arms. Typically, the body below the arms is raised by about 30 cm in a time of 1.0 s, starting from rest. Assume that the entire body of a 700 N person doing chin-ups is raised by 30 cm, and that half the 1.0 s is spent accelerating upward and the other half accelerating downward, uniformly in both cases. Draw a free-body diagram of the person's body, and use it to find the force his arms must exert on him during the accelerating part of the chin-up.

5.68 •• **CP** **CALC** A 2.00 kg box is suspended from the end of a light vertical rope. A time-dependent force is applied to the upper end of the rope, and the box moves upward with a velocity magnitude that varies in time according to $v(t) = (2.00\ \text{m/s}^2)t + (0.600\ \text{m/s}^3)t^2$. What is the tension in the rope when the velocity of the box is 9.00 m/s?

5.69 ••• **CALC** A 3.00 kg box that is several hundred meters above the earth's surface is suspended from the end of a short vertical rope of negligible mass. A time-dependent upward force is applied to the upper end of the rope and results in a tension in the rope of $T(t) = (36.0\ \text{N/s})t$. The box is at rest at $t = 0$. The only forces on the box are the tension in the rope and gravity. (a) What is the velocity of the box at (i) $t = 1.00$ s and (ii) $t = 3.00$ s? (b) What is the maximum distance that the box descends below its initial position? (c) At what value of t does the box return to its initial position?

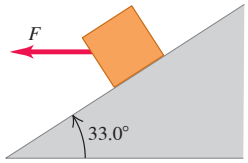
5.70 •• **CP** A 5.96 kg box sits at rest at the bottom of a ramp that is 8.00 m long and is inclined at 35.6° above the horizontal. The coefficient of kinetic friction is $\mu_k = 0.23$, and the coefficient of static friction is $\mu_s = 0.43$. What constant force F, applied parallel to the surface of the ramp, is required to push the box to the top of the ramp in a time of 6.00 s?

5.71 •• When a crate with mass 25.0 kg is placed on a ramp that is inclined at an angle α below the horizontal, it slides down the ramp with an acceleration of 4.9 m/s^2. The ramp is not frictionless. To increase the acceleration of the crate, a downward vertical force $\vec{F}$ is applied to the top of the crate. What must F be in order to increase the acceleration of the crate so that it is 9.8 m/s^2? How does the value of F that you calculate compare to the weight of the crate?

5.72 •• A large crate is at rest on a horizontal floor. The coefficient of static friction between the crate and the floor is 0.400. A force $\vec{F}$ is applied to the crate in a direction 30.0° above the horizontal. The minimum value of F required to get the crate to start sliding is 380 N. What is the mass of the crate?

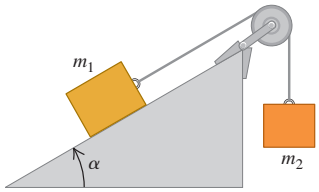
5.73 •• **CP** An 8.00 kg box sits on a ramp that is inclined at 33.0° above the horizontal. The coefficient of kinetic friction between the box and the surface of the ramp is $\mu_k = 0.300$. A constant *horizontal* force $F = 26.0$ N is applied to the box (**Fig. P5.73**), and the box moves down the ramp. If the box is initially at rest, what is its speed 2.00 s after the force is applied?

Figure **P5.73**



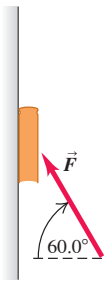
5.74 •• **CP** In **Fig. P5.74,** $m_1 = 20.0$ kg and $\alpha = 53.1°$. The coefficient of kinetic friction between the block of mass m_1 and the incline is $\mu_k = 0.40$. What must be the mass m_2 of the hanging block if it is to descend 12.0 m in the first 3.00 s after the system is released from rest?

Figure **P5.74**



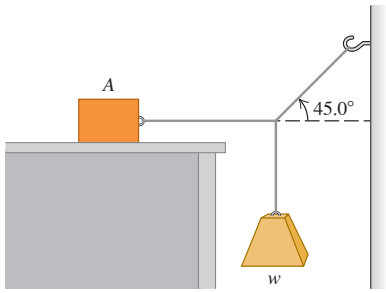
5.75 •• **CP** You place a book of mass 5.00 kg against a vertical wall. You apply a constant force $\vec{F}$ to the book, where $F = 96.0$ N and the force is at an angle of 60.0° above the horizontal (**Fig. P5.75**). The coefficient of kinetic friction between the book and the wall is 0.300. If the book is initially at rest, what is its speed after it has traveled 0.400 m up the wall?

Figure **P5.75**



5.76 •• Block A in **Fig. P5.76** weighs 64.5 N. The coefficient of static friction between the block and the surface on which it rests is 0.30. The weight w is 11.4 N, and the system is in equilibrium. (a) Find the friction force exerted on block A. (b) Find the maximum weight w for which the system will remain in equilibrium.

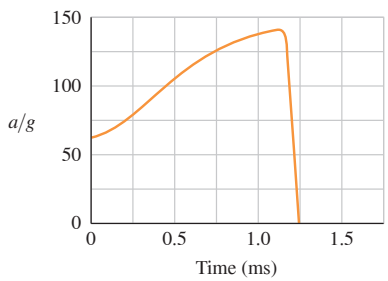
Figure **P5.76**



5.77 •• A block with mass m_1 is placed on an inclined plane with slope angle α and is connected to a hanging block with mass m_2 by a cord passing over a small, frictionless pulley (Fig. P5.74). The coefficient of static friction is μ_s, and the coefficient of kinetic friction is μ_k. (a) Find the value of m_2 for which the block of mass m_1 moves up the plane at constant speed once it is set in motion. (b) Find the value of m_2 for which the block of mass m_1 moves down the plane at constant speed once it is set in motion. (c) For what range of values of m_2 will the blocks remain at rest if they are released from rest?

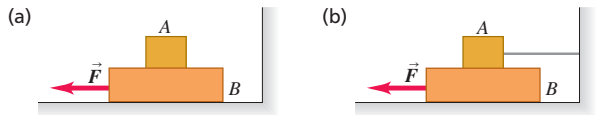
5.78 •• DATA BIO **The Flying Leap of a Flea.** High-speed motion pictures (3500 frames/second) of a jumping 210 μg flea yielded the data to plot the flea's acceleration as a function of time, as shown in **Fig. P5.78.** (See "The Flying Leap of the Flea," by M. Rothschild et al., *Scientific American,* November 1973.) This flea was about 2 mm long and jumped at a nearly vertical takeoff angle. Using the graph, (a) find the *initial* net external force on the flea. How does it compare to the flea's weight? (b) Find the *maximum* net external force on this jumping flea. When does this maximum force occur? (c) Use the graph to find the flea's maximum speed.

Figure P5.78



5.79 •• Block A in **Fig. P5.79** weighs 1.20 N, and block B weighs 3.60 N. The coefficient of kinetic friction between all surfaces is 0.300. Find the magnitude of the horizontal force $\vec{F}$ necessary to drag block B to the left at constant speed (a) if A rests on B and moves with it (Fig. P5.79a), (b) if A is held at rest (Fig. P5.79b).

Figure P5.79



5.80 ••• CP **Lift Design.** You are designing an lift for a hospital. The force exerted on a passenger by the floor of the lift is not to exceed 1.60 times the passenger's weight. The lift accelerates upward with constant acceleration for a distance of 3.0 m and then starts to slow down. What is the maximum speed of the lift?

5.81 ••• CP CALC You are standing on a bathroom scale a lift in a tall building. Your mass is 64 kg. The lift starts from rest and travels upward with a speed that varies with time according to $v(t) = (3.0\ \text{m/s}^2)t + (0.20\ \text{m/s}^3)t^2$. When $t = 4.0$ s, what is the reading on the bathroom scale?

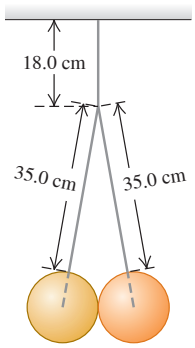
5.82 •• CP Consider the system shown in Fig. E5.34. As in Exercise 5.34, block A weighs 45.0 N and block B weighs 25.0 N. The system is released from rest, and block B has speed 3.30 m/s after it has descended 2.00 m. (a) While the blocks are moving, what is the tension in the rope connecting the blocks? (b) What is the coefficient of kinetic friction between block A and the tabletop?

5.83 •• CP Two blocks are suspended from opposite ends of a light rope that passes over a light, frictionless pulley. One block has mass m_1 and the other has mass m_2, where $m_2 > m_1$. The two blocks are released from rest, and the block with mass m_2 moves downward 5.00 m in 2.00 s after being released. While the blocks are moving, the tension in the rope is 16.0 N. Calculate m_1 and m_2.

5.84 ••• If the coefficient of static friction between a table and a uniform, massive rope is μ_s, what fraction of the rope can hang over the edge of the table without the rope sliding?

5.85 ••• Two identical 16.1 kg balls, each 21.1 cm in diameter, are suspended by two 35.0 cm wires (**Fig. P5.85**). The entire apparatus is supported by a single 18.0 cm wire, and the surfaces of the balls are perfectly smooth. (a) Find the tension in each of the three wires. (b) How hard does each ball push on the other one?

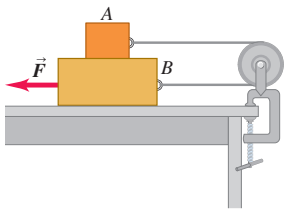
Figure P5.85



5.86 • CP **Traffic Court.** You are called as an expert witness in a trial for a traffic violation. The facts are these: A driver slammed on his brakes and came to a stop with constant acceleration. Measurements of his tires and the skid marks on the pavement indicate that he locked his car's wheels, the car traveled 23.3 m before stopping, and the coefficient of kinetic friction between the road and his tires was 0.750. He was charged with speeding in a 50 km/h zone but pleads innocent. What is your conclusion: guilty or innocent? How fast was he going when he hit his brakes?

5.87 ••• Block A in **Fig. P5.87** weighs 1.40 N, and block B weighs 3.66 N. The coefficient of kinetic friction between all surfaces is 0.36. Find the magnitude of the horizontal force $\vec{F}$ necessary to drag block B to the left at constant speed if A and B are connected by a light, flexible cord passing around a fixed, frictionless pulley.

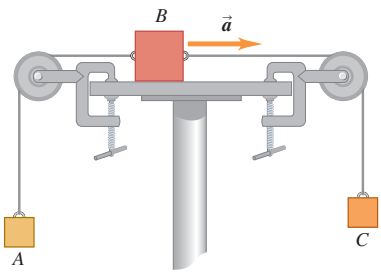
Figure P5.87



5.88 •• Block B has mass 5.00 kg and sits at rest on a horizontal, frictionless surface. Block A has mass 2.00 kg and sits at rest on top of block B. The coefficient of static friction between the two blocks is 0.400. A horizontal force $\vec{P}$ is then applied to block A. What is the largest value P can have and the blocks move together with equal accelerations?

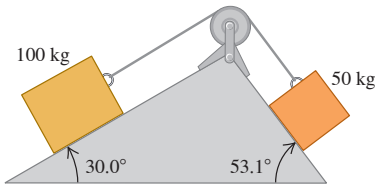
5.89 •• Block A in **Fig. P5.89** has mass 5.00 kg, and block B has mass 14.0 kg. The coefficient of kinetic friction between block B and the horizontal surface is 0.25. (a) What is the mass of block C if block B is moving to the right and speeding up with an acceleration of 2.70 m/s^2? (b) What is the tension in each cord when block B has this acceleration?

Figure P5.89



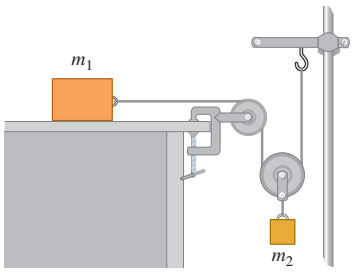
5.90 •• Two blocks connected by a cord passing over a small, frictionless pulley rest on frictionless planes (**Fig. P5.90**). (a) Which way will the system move when the blocks are released from rest? (b) What is the acceleration of the blocks? (c) What is the tension in the cord?

Figure P5.90



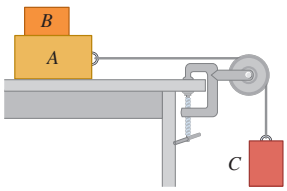
5.91 •• In terms of m_1, m_2, and g, find the acceleration of each block in **Fig. P5.91.** There is no friction anywhere in the system.

Figure P5.91



5.92 ••• Block B, with mass 5.00 kg, rests on block A, with mass 8.00 kg, which in turn is on a horizontal tabletop (**Fig. P5.92**). There is no friction between block A and the tabletop, but the coefficient of static friction between blocks A and B is 0.750. A light string attached to block A passes over a frictionless, massless pulley, and block C is suspended from the other end of the string. What is the largest mass that block C can have so that blocks A and B still slide together when the system is released from rest?

Figure P5.92

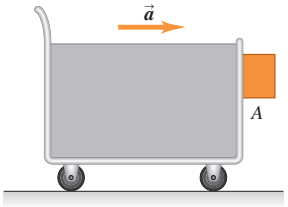


5.93 •• Consider two blocks connected as shown in Fig. P5.87. Block A has mass 2.00 kg, and block B has mass 5.00 kg. The table on which B sits is frictionless, the cord connecting the blocks is light and flexible, and the pulley is light and frictionless. The horizontal force $\vec{F}$ has magnitude $F = 20.0$ N, and block B moves to the left with an acceleration of 1.50 m/s^2. (a) What is the tension in the cord that connects the two blocks? (b) What is the coefficient of kinetic friction that one block exerts on the other?

5.94 •• **Friction in a Lift.** You are riding in a lift on the way to the 18th floor of your dormitory. The lift is accelerating upward with $a = 1.81$ m/s^2. Beside you is the box containing your new computer; the box and its contents have a total mass of 29.0 kg. While the lift is accelerating upward, you push horizontally on the box to slide it at constant speed toward the lift door. If the coefficient of kinetic friction between the box and the lift floor is $\mu_k = 0.31$, what magnitude of force must you apply?

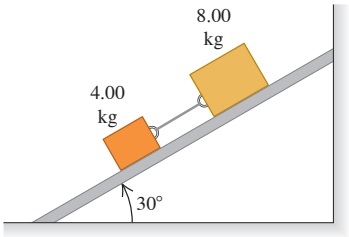
5.95 • A block is placed against the vertical front of a cart (**Fig. P5.95**). What acceleration must the cart have so that block A does not fall? The coefficient of static friction between the block and the cart is μ_s. How would an observer on the cart describe the behavior of the block?

Figure P5.95



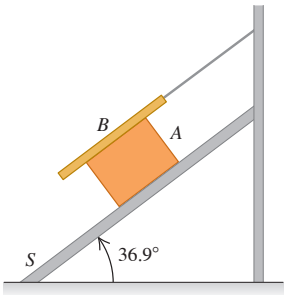
5.96 ••• Two blocks, with masses 4.00 kg and 8.00 kg, are connected by a string and slide down a 30.0° inclined plane (**Fig. P5.96**). The coefficient of kinetic friction between the 4.00 kg block and the plane is 0.25; that between the 8.00 kg block and the plane is 0.35. Calculate (a) the acceleration of each block and (b) the tension in the string. (c) What happens if the positions of the blocks are reversed, so that the 4.00 kg block is uphill from the 8.00 kg block?

Figure P5.96



5.97 ••• Block A, with weight $3w$, slides down an inclined plane S of slope angle 36.9° at a constant speed while plank B, with weight w, rests on top of A. The plank is attached by a cord to the wall (**Fig. P5.97**). (a) Draw a diagram of all the forces acting on block A. (b) If the coefficient of kinetic friction is the same between A and B and between S and A, determine its value.

Figure P5.97



5.98 •• Jack sits in the chair of a Ferris wheel that is rotating at a constant 0.100 rev/s. As Jack passes through the highest point of his circular path, the upward force that the chair exerts on him is equal to one-fourth of his weight. What is the radius of the circle in which Jack travels? Treat him as a point mass.

5.99 •• On the now defunct ride "Spindletop" at the amusement park Six Flags Over Texas, people stood against the inner wall of a hollow vertical cylinder with radius 2.5 m. The cylinder started to rotate, and when it reached a constant rotation rate of 0.60 rev/s, the floor dropped about 0.5 m. The people remained pinned against the wall without touching the floor. (a) Draw a force diagram for a person on this ride after the floor has dropped. (b) What minimum coefficient of static friction was required for the person not to slide downward to the new position of the floor? (c) Does your answer in part (b) depend on the person's mass? (*Note:* When such a ride is over, the cylinder is slowly brought to rest. As it slows down, people slide down the wall to the floor.)

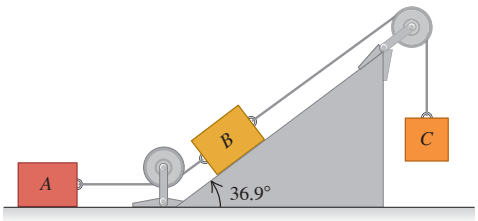
5.100 •• A large piece of ice breaks loose from the roof of a chalet at a ski resort and slides across the snow-covered terrain. It passes over the top of a hill in a path that has the shape of the arc of a circle of radius R. What is the speed v of the piece of ice as it passes over the top of the hill if the normal force exerted on it at this point is half its weight?

5.101 ••• A racetrack curve has radius 90.0 m and is banked at an angle of 18.0°. The coefficient of static friction between the tires and the roadway is 0.400. A race car with mass 1200 kg rounds the curve with the maximum speed to avoid skidding. (a) As the car rounds the curve, what is the normal force exerted on it by the road? What are the car's (b) radial acceleration and (c) speed?

5.102 ••• A racetrack curve has radius 120.0 m and is banked at an angle of 18.0°. The coefficient of static friction between the tires and the roadway is 0.300. A race car with mass 900 kg rounds the curve with the minimum speed needed to not slide down the banking. (a) As the car rounds the curve, what is the normal force exerted on it by the road? (b) What is the car's speed?

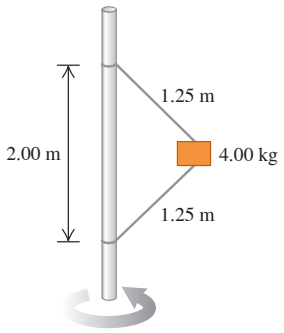
5.103 ••• Blocks A, B, and C are placed as in **Fig. P5.103** and connected by ropes of negligible mass. Both A and B weigh 23.8 N each, and the coefficient of kinetic friction between each block and the surface is 0.32. Block C descends with constant velocity. (a) Draw separate free-body diagrams showing the forces acting on A and on B. (b) Find the tension in the rope connecting blocks A and B. (c) What is the weight of block C? (d) If the rope connecting A and B were cut, what would be the acceleration of C?

Figure **P5.103**



5.104 ••• A 4.00 kg block is attached to a vertical rod by means of two strings. When the system rotates about the axis of the rod, the strings are extended as shown in **Fig. P5.104** and the tension in the upper string is 80.0 N. (a) What is the tension in the lower cord? (b) How many revolutions per minute does the system make? (c) Find the number of revolutions per minute at which the lower cord just goes slack. (d) Explain what happens if the number of revolutions per minute is less than that in part (c).

Figure **P5.104**

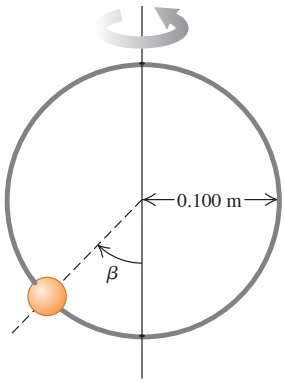


5.105 •• **CALC** You throw a rock downward into water with a speed of $3mg/k$, where k is the coefficient in Eq. (5.5). Assume that the relationship between fluid resistance and speed is as given in Eq. (5.5), and calculate the speed of the rock as a function of time.

5.106 ••• A box with mass m sits at the bottom of a long ramp that is sloped upward at an angle α above the horizontal. You give the box a quick shove, and after it leaves your hands it is moving up the ramp with an initial speed v_0. The box travels a distance d up the ramp and then slides back down. When it returns to its starting point, the speed of the box is half the speed it started with; it has speed $v_0/2$. What is the coefficient of kinetic friction between the box and the ramp? (Your answer should depend on only α.)

5.107 •• A small bead can slide without friction on a circular hoop that is in a vertical plane and has a radius of 0.100 m. The hoop rotates at a constant rate of 4.80 rev/s about a vertical diameter (**Fig. P5.107**). (a) Find the angle β at which the bead is in vertical equilibrium. (It has a radial acceleration toward the axis.) (b) Is it possible for the bead to "ride" at the same elevation as the center of the hoop? (c) What will happen if the hoop rotates at 1.00 rev/s?

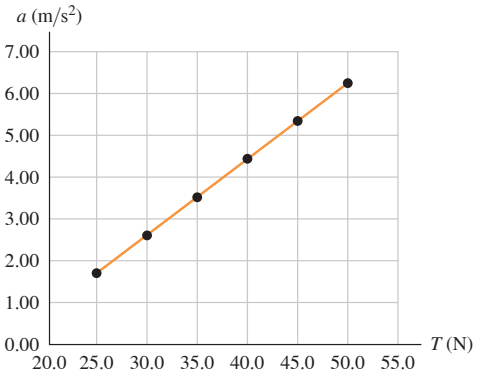
Figure **P5.107**



5.108 •• Gina works in a traveling funfair. She rides a motorcycle inside a hollow, transparent plastic sphere. After gaining sufficient speed, she travels in a vertical circle with radius 14.2 m. She has mass 68.0 kg, and her motorcycle has mass 40.0 kg. (a) What minimum speed must she have at the top of the circle for the motorcycle tires to remain in contact with the sphere? (b) At the bottom of the circle, her speed is twice the value calculated in part (a). What is the magnitude of the normal force exerted on the motorcycle by the sphere at this point?

5.109 •• **DATA** In your physics lab, a block of mass m is at rest on a horizontal surface. You attach a light cord to the block and apply a horizontal force to the free end of the cord. You find that the block remains at rest until the tension T in the cord exceeds 20.0 N. For $T > 20.0$ N, you measure the acceleration of the block when T is maintained at a constant value, and you plot the results (**Fig. P5.109**). The equation for the straight line that best fits your data is $a = [0.182\ \text{m}/(\text{N}\cdot\text{s}^2)]T - 2.842\ \text{m}/\text{s}^2$. For this block and surface, what are (a) the coefficient of static friction and (b) the coefficient of kinetic friction? (c) If the experiment were done on the earth's moon, where g is much smaller than on the earth, would the graph of a versus T still be fit well by a straight line? If so, how would the slope and intercept of the line differ from the values in Fig. P5.109? Or, would each of them be the same?

Figure **P5.109**



5.110 •• **DATA** A road heading due east passes over a small hill. You drive a car of mass m at constant speed v over the top of the hill, where the shape of the roadway is well approximated as an arc of a circle with radius R. Sensors have been placed on the road surface there to measure the downward force that cars exert on the surface at various speeds. The table gives values of this force versus speed for your car:

Speed (m/s)	6.00	8.00	10.0	12.0	14.0	16.0
Force (N)	8100	7690	7050	6100	5200	4200

Treat the car as a particle. (a) Plot the values in such a way that they are well fitted by a straight line. You might need to raise the speed, the force, or both to some power. (b) Use your graph from part (a) to calculate m and R. (c) What maximum speed can the car have at the top of the hill and still not lose contact with the road?

5.111 •• **DATA** You are an engineer working for a manufacturing company. You are designing a mechanism that uses a cable to drag heavy metal blocks a distance of 8.00 m along a ramp that is sloped at 40.0° above the horizontal. The coefficient of kinetic friction between these blocks and the incline is $\mu_k = 0.350$. Each block has a mass of 2170 kg. The block will be placed on the bottom of the ramp, the cable will be attached, and the block will then be given just enough of a momentary push to overcome static friction. The block is then to accelerate at a constant rate to move the 8.00 m in 4.20 s. The cable is made of wire rope and is parallel to the ramp surface. The table gives the breaking strength of the cable as a function of its diameter; the safe load tension, which is 20% of the breaking strength; and the mass per meter of the cable:

Cable Diameter (mm)	**Breaking Strength (kN)**	**Safe Load (kN)**	**Mass per Meter (kg/m)**
6.4	24.4	4.89	0.16
9.5	54.3	10.9	0.36
13	95.2	19.0	0.63
16	149	29.7	0.98
19	212	42.3	1.41
22	286	57.4	1.92
26	372	74.3	2.50

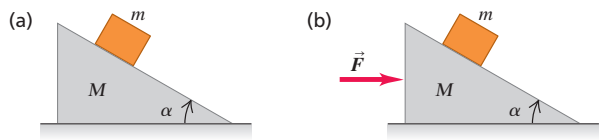
Source: www.engineeringtoolbox.com

(a) What is the minimum diameter of the cable that can be used to pull a block up the ramp without exceeding the safe load value of the tension in the cable? Ignore the mass of the cable, and select the diameter from those listed in the table. (b) You need to know safe load values for diameters that aren't in the table, so you hypothesize that the breaking strength and safe load limit are proportional to the cross-sectional area of the cable. Draw a graph that tests this hypothesis, and discuss its accuracy. What is your estimate of the safe load value for a cable with diameter 14.5 mm? (c) The coefficient of static friction between the crate and the ramp is $\mu_s = 0.620$, which is nearly twice the value of the coefficient of kinetic friction. If the machinery jams and the block stops in the middle of the ramp, what is the tension in the cable? Is it larger or smaller than the value when the block is moving? (d) Is the actual tension in the cable, at its upper end, larger or smaller than the value calculated when you ignore the mass of the cable? If the cable is 9.00 m long, how accurate is it to ignore the cable's mass?

CHALLENGE PROBLEMS

5.112 ••• **Moving Wedge.** A wedge with mass M rests on a frictionless, horizontal tabletop. A block with mass m is placed on the wedge (**Fig. P5.112a**). There is no friction between the block and the wedge. The system is released from rest. (a) Calculate the acceleration of the wedge and the horizontal and vertical components of the acceleration of the block. (b) Do your answers to part (a) reduce to the correct results when M is very large? (c) As seen by a stationary observer, what is the shape of the trajectory of the block?

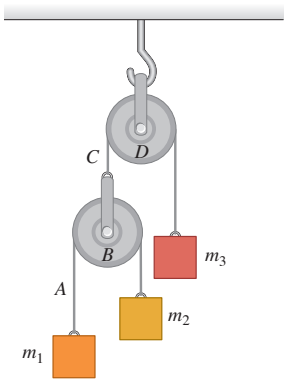
Figure **P5.112**



5.113 ••• A wedge with mass M rests on a frictionless, horizontal tabletop. A block with mass m is placed on the wedge, and a horizontal force $\vec{F}$ is applied to the wedge (Fig. P5.112b). What must the magnitude of $\vec{F}$ be if the block is to remain at a constant height above the tabletop?

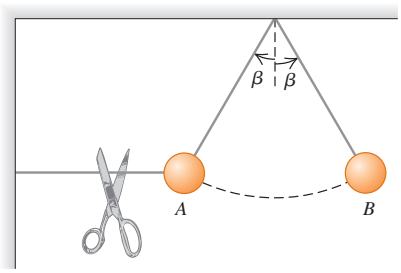
5.114 ••• **Double Atwood's Machine.** In **Fig. P5.114** masses m_1 and m_2 are connected by a light string A over a light, frictionless pulley B. The axle of pulley B is connected by a light string C over a light, frictionless pulley D to a mass m_3. Pulley D is suspended from the ceiling by an attachment to its axle. The system is released from rest. In terms of m_1, m_2, m_3, and g, what are (a) the acceleration of block m_3; (b) the acceleration of pulley B; (c) the acceleration of block m_1; (d) the acceleration of block m_2; (e) the tension in string A; (f) the tension in string C? (g) What do your expressions give for the special case of $m_1 = m_2$ and $m_3 = m_1 + m_2$? Is this reasonable?

Figure P5.114



5.115 ••• A ball is held at rest at position A in **Fig. P5.115** by two light strings. The horizontal string is cut, and the ball starts swinging as a pendulum. Position B is the farthest to the right that the ball can go as it swings back and forth. What is the ratio of the tension in the supporting string at B to its value at A before the string was cut?

Figure P5.115



MCAT-STYLE PASSAGE PROBLEMS

Friction and Climbing Shoes. Shoes made for the sports of bouldering and rock climbing are designed to provide a great deal of friction between the foot and the surface of the ground. Such shoes on smooth rock might have a coefficient of static friction of 1.2 and a coefficient of kinetic friction of 0.90.

5.116 For a person wearing these shoes, what's the maximum angle (with respect to the horizontal) of a smooth rock that can be walked on without slipping? (a) 42°; (b) 50°; (c) 64°; (d) larger than 90°.

5.117 If the person steps onto a smooth rock surface that's inclined at an angle large enough that these shoes begin to slip, what will happen? (a) She will slide a short distance and stop; (b) she will accelerate down the surface; (c) she will slide down the surface at constant speed; (d) we can't tell what will happen without knowing her mass.

5.118 A person wearing these shoes stands on a smooth, horizontal rock. She pushes against the ground to begin running. What is the maximum horizontal acceleration she can have without slipping? (a) $0.20g$; (b) $0.75g$; (c) $0.90g$; (d) $1.2g$.

ANSWERS

Chapter Opening Question ?

(iii) The upward force exerted by the air has the same magnitude as the force of gravity. Although the seed and pappus are descending, their vertical velocity is constant, so their vertical acceleration is zero. According to Newton's first law, the net vertical force on the seed and pappus must also be zero. The individual vertical forces must balance.

Key Example VARIATION Problems

VP5.5.1 (a) 425 N (b) 255 N
VP5.5.2 (a) 367 N (b) 522 N
VP5.5.3 (a) 21.8° (b) 637 N
VP5.5.4 $w(1 - \sin\theta)$, less than w
VP5.15.1 (a) 79.3 N (b) 298 N
VP5.15.2 (a) 90.2 N (b) 348 N
VP5.15.3 (a) 449 N (b) 0.670 m/s^2, speeding up
VP5.15.4 $\mu_s = (T_{\min}\cos\theta)/(mg - T_{\min}\sin\theta)$
VP5.22.1 (a) 0.274 m (b) 1.74 s (c) 2.61 N
VP5.22.2 (a) 19.0 m (b) 8.22 m/s^2 (c) 819 N
VP5.22.3 (a) 75.0° (b) 3.03×10^3 N, 3.86 times greater
VP5.22.4 (a) $mg\sqrt{x^2 - 1}$ (b) $\dfrac{v^2}{g\sqrt{x^2 - 1}}$

Bridging Problem

(a) $T_{\max} = 2\pi\sqrt{\dfrac{h(\cos\beta + \mu_s\sin\beta)}{g\tan\beta(\sin\beta - \mu_s\cos\beta)}}$

(b) $T_{\min} = 2\pi\sqrt{\dfrac{h(\cos\beta - \mu_s\sin\beta)}{g\tan\beta(\sin\beta + \mu_s\cos\beta)}}$